



FIT3152 – DATA ANALYTICS A3

Natural Language Processing & Network Analysis



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Larana, Inc.

INTRODUCTION

- A corpus of **20** documents
- **4** genres - Music, News, Recipe, Review
- **Tokens (DTM) - 26** after sparsity reduction
- Text preprocessing, sentiment analysis, hierarchical clustering & network analysis

CORPUS & TEXT PREPROCESSING

Genres collected

- Music - BTS comeback, tour, interviews
- News - Social media law, road bully, passport
- Recipe - Ramen, Kung Pao chicken, dumplings
- Review - Horror, Zombie, K-drama, Films

Preprocessing steps

- Lowercase, remove URLs & numbers
- Remove punctuation & stopwords
- Stem words to root forms
- Sparsity reduction at 0.65 threshold
- 3,038 terms → 26 retained tokens

Key retained tokens - make, time, work, use first, year, said, social, feel world

HIERARCHICAL CLUSTERING - 75% ACCURACY

Results by genre

Recipe

100%

Review

100%

News

60%

Music

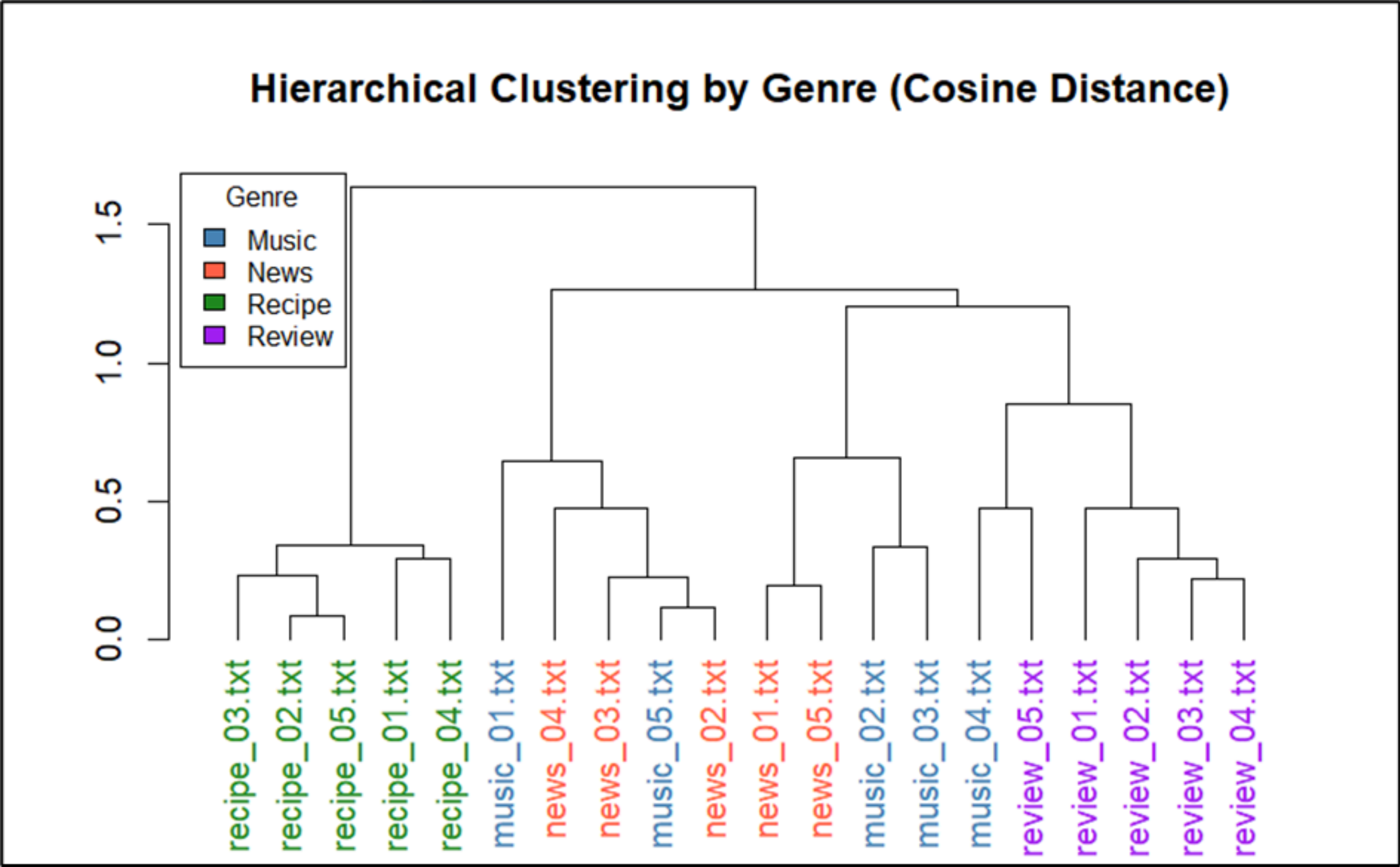
40%

Why Music & News overlap

- Music documents are news-style articles about BTS tours and artist releases
- Both genres share journalistic terms: said, year, new, first
- Cosine distance cannot separate them when vocabulary is nearly identical

Method - cosine distance + Ward's linkage, cut at $k = 4$

HIERARCHICAL CLUSTERING - 75% ACCURACY



SENTIMENT ANALYSIS - GI DICTIONARY

Music mean
SentimentGI
100%

News mean
SentimentGI
100%

Recipe mean
SentimentGI
60%

Review mean
SentimentGI
40%

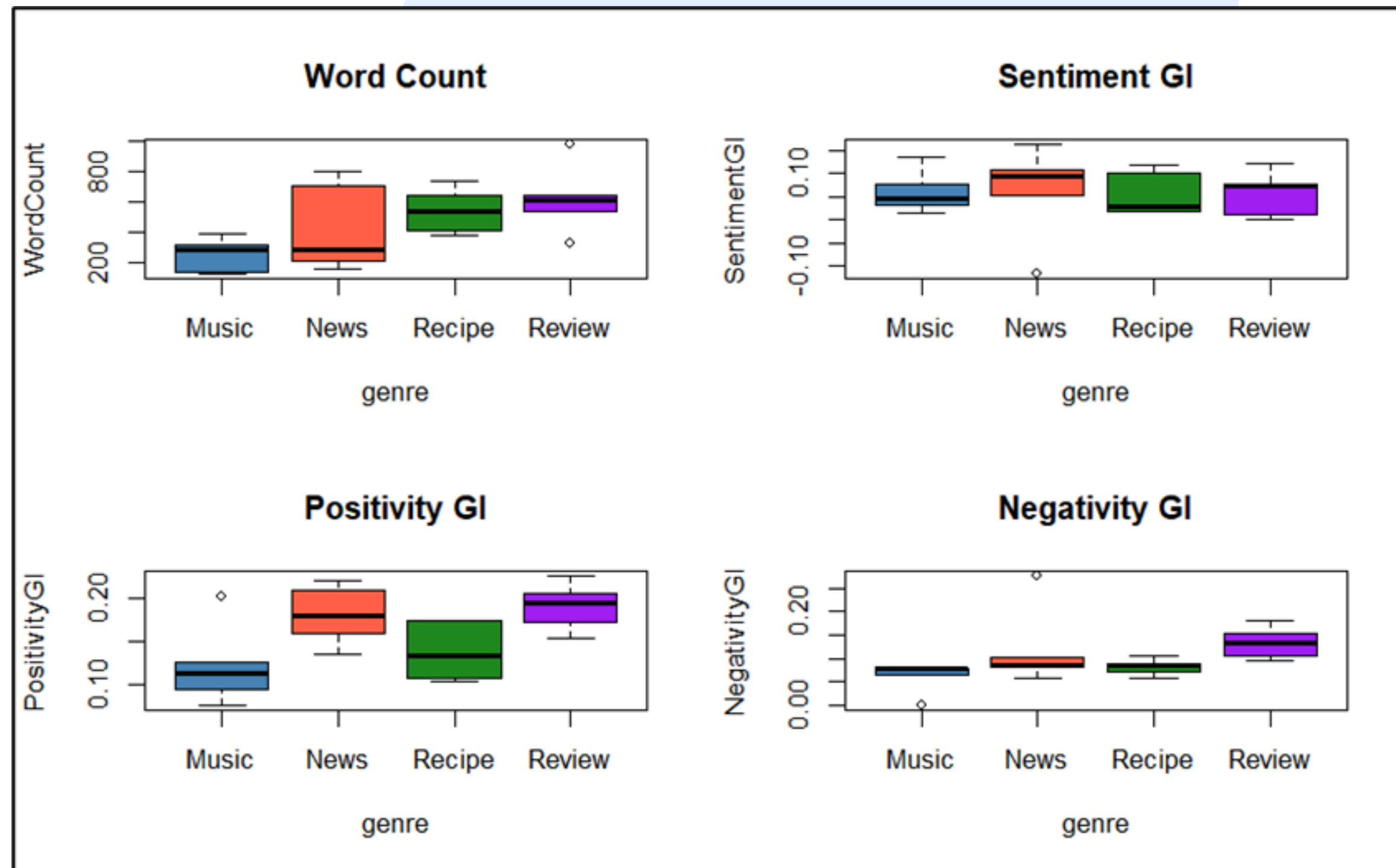
Key findings

- All genres have near-identical net sentiment
- ANOVA: $F = 0.008$, $p = 0.999$ — not significant
- Reviews have highest positivity (0.190) AND negativity (0.134)
- News has highest variability ($SD = 0.106$)

Outlier

- **news03**, only negative document (SentimentGI = -0.116).
- Coverage of a road bully conviction case.

SENTIMENT ANALYSIS GI DICTIONARY



6 DOCUMENT NETWORK

Network Structure

- 20 nodes (documents), edges = shared tokens
- Fully connected — 17/20 documents at max degree 19
- Louvain found 2 communities (45% accuracy)

Communities

Community 1

News + music_01, music_05

Community 2

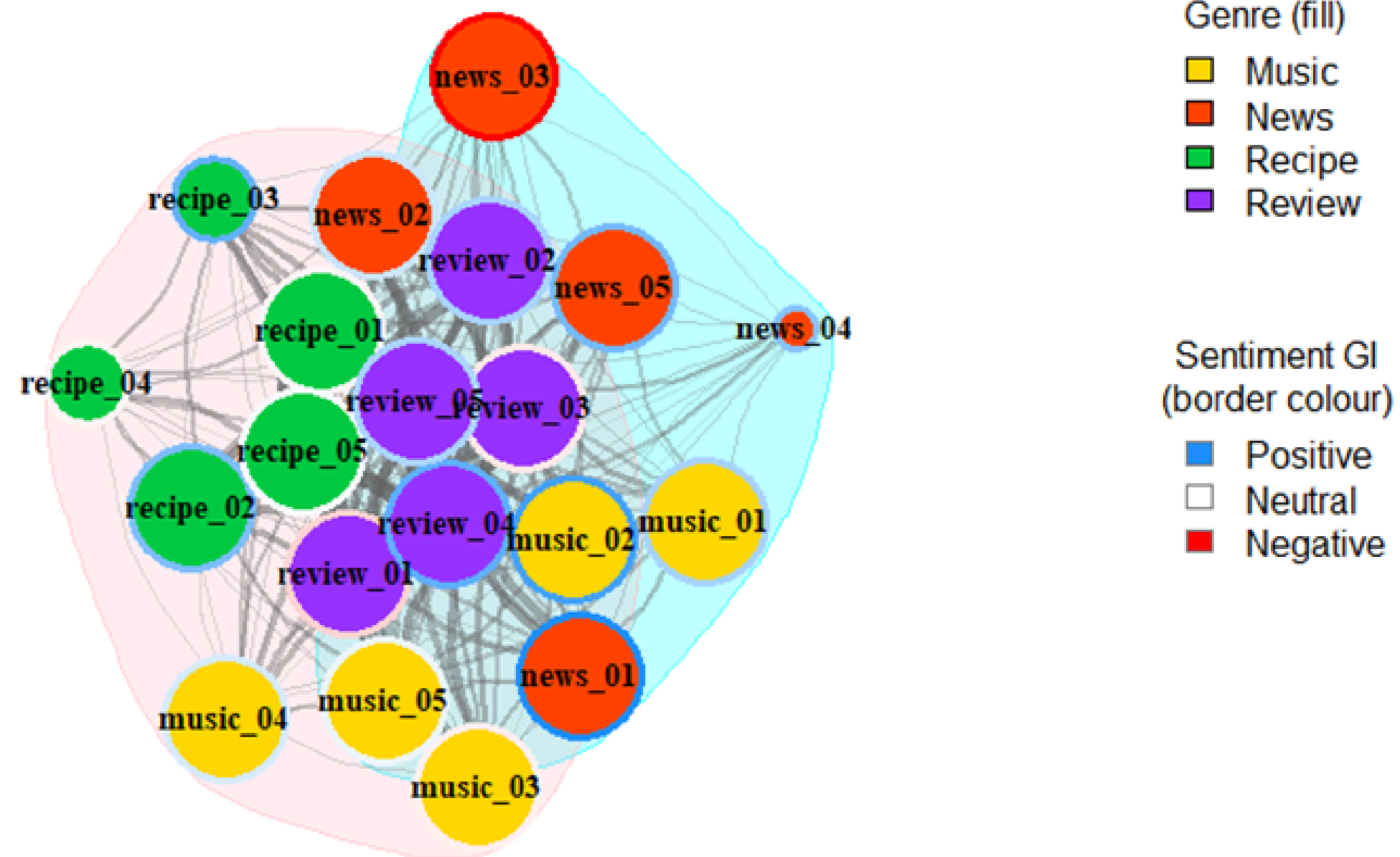
All Recipe + Review + music_02-04, news_02

Central documents

Highest betweenness (bridge)
news_04 (0.361)
Connects the two communities

Highest eigenvector (core)
review_04 (1.000)
Most influential, embedded in dense cluster

6 DOCUMENT NETWORK



7 TOKEN NETWORK

Network Structure

- 20 nodes (documents), edges = shared tokens
- Fully connected — 17/20 documents at max degree 19
- Louvain found 2 communities (45% accuracy)

Communities

Community 1 (Recipe/Review language)

make, use, work, take, come, well

Community 2 (Music/News language)

said, year, new, social, world, first...

Central tokens

Highest eigenvector (core)

make (1.000)

Appears in 16/20 docs, co-occurs with every token

Highest betweenness (bridge)

social (0.122)

Links the two vocabulary communities



8 BIPARTITE NETWORK

Network Structure

- 46 nodes: 20 documents + 26 tokens
- 254 edges — document to token only
- No document–document or token–token edges

Top document pairs (shared tokens)

review_04 – review_05 17 shared

recipe_05 – review_04 16 shared

All top 10 pairs involve only Recipe or Review docs

Most central nodes

Most connected document

review_04 (deg 22)

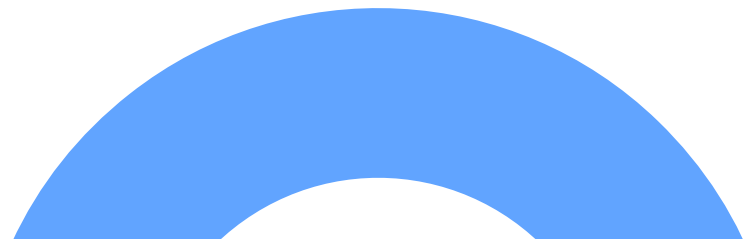
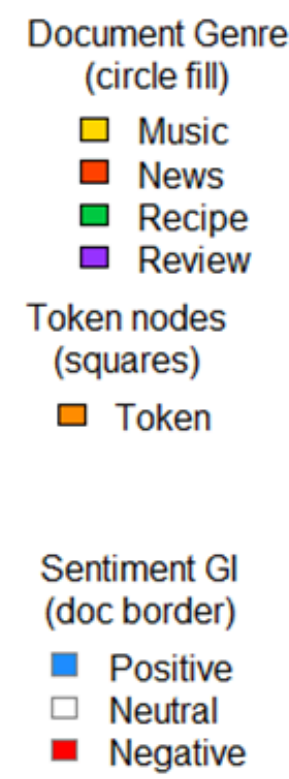
Most connected token

make (16 docs)

Most peripheral document

news_04 (deg 5)

Figure 8.2 – Bipartite Network (Improved)



KEY FINDINGS & CLUSTERING VS NETWORKS

Consistent findings across all analyses

- **Recipe + Review** always cluster together — shared instructional/evaluative language
- **Music + News** overlap due to news-style articles in the Music corpus
- **review_04** is the most central document overall
- **make** is the most central token overall
- **news_03** only negative sentiment document

Clustering vs networks

Clustering accuracy

75%

Best for genre classification

Network community accuracy

45%

Better for relationships & importance

Networks revealed why documents cluster — through token communities and centrality measures that clustering cannot provide.

SUGGESTED IMPROVEMENTS TO TEXT PROCESSING

Bag-of-words DTM ignores word order, context, and meaning

TF-IDF

Better DTM weights. Down-weight common cross-genre terms, up-weight distinctive ones

Word embeddings — Word2Vec or GloVe

Semantic similarity beyond exact vocabulary (Separate Music from News)

BERT

- Context-sensitive representations distinguish same word in different genres
- Highest accuracy & most computationally costly



THANK YOU



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WHY DOES DATA PRIVACY MATTER?

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TYPES OF PERSONAL DATA

Basic Data

- Name
- Address
- Phone number

Sensitive Data

- Health records
- Political opinions
- Biometric data

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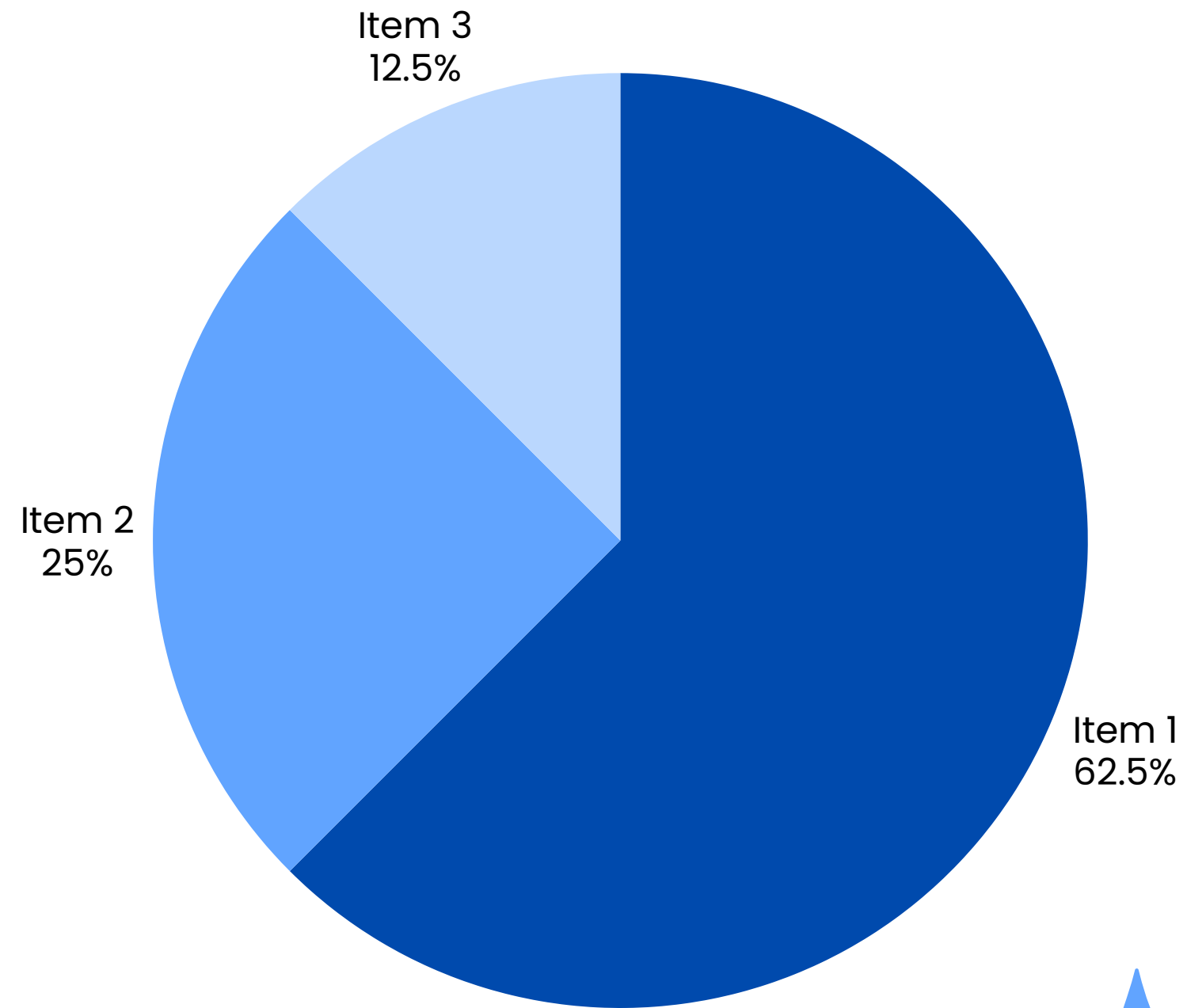
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COMMON PRIVACY THREATS

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HOW VULNERABLE ARE WE?

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TYPES OF PERSONAL DATA

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Basic Data

- Sharing real-time location
- Photos with sensitive info in background
- Accepting unknown friend requests

Sensitive Data

- Post later or disable location
- Review background before posting
- Connect only with known people

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YOUR DATA RIGHTS

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THE ROLE OF DIGITAL EDUCATION

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